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Biofoulings on biofoulings: Analysis of relationship between fouling behaviors of organisms in Shenzhen Bay

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Abstract

In the marine ecosystem, many foulers and benthic organisms settle not only on the inorganic surface but also on other living foulers. Certain species of foulers may favor different surface components to attach to. In this experiment, biofouling samples that are attached to other foulers are collected in Dapeng (eastern part of Shenzhen), and photo and rough classification are recorded. This is followed by a PCR test on the genetic sequence of the mitochondrial gene cytochrome oxidase I (COI). The results species identification are then compared and contrasted with the rough classification to identify the final classification. The result demonstrates a pattern of fouling on certain species of foulers. The result also reveals a phenomenon that contributed to the differences between PCR classification and rough classification.

1. Introduction

Biofouling organisms refer to animals or plants that tend to settle and morph on natural or artificial surfaces. Biofouling communities are widely seen in all marine and freshwater regions. The feature of its rapid expansion and fouling makes biofouling communities a major threat to human infrastructure, facilities, and vehicles. Some fouling organisms such as barnacles, bivalves, and bryozoan secrete glue substances that assist them to settle on the surface and also produce a calcified skeleton/exoskeleton. This characteristic leads to biofouling communities hard to be removed and damage the artificial surface. Thus the studies on the behavior and composition of biofouling communities are important for the development of inhibition and control of biofouling communities.

Based on their settlement, biofouling organisms can be classified as “foulers” and “foulers of foulers”. Foulers are organisms that initially settle on the surface directly. While “foulers of foulers” are organisms that settle on fouling organisms. Multiple reasons led to the existence of “foulers of foulers”, including competition for space and area, forming communities of the same species for reproduction, and ensuring the stability of fouler itself. While biofouling communities have immense complexity in all three dimensions, little research was conducted on the behavior and characteristics of “foulers of foulers”. Understanding the features of “foulers of foulers” helps the analysis of biofouling communities in terms of their composition, their ability to spread, and their damage to the fouling surface.

This experiment aims to study the ecological niche of “foulers of foulers”, their impact on biofouling communities, and their relationship with other foulers through sampling and species identification by Sanger sequencing of the mitochondrial gene cytochrome oxidase I (COI). Sample images were taken and their relationship with other foulers in the fouling community was recorded, and the result of sequencing was compared to the rough classification based on the image taken.

2. Materials and Methods

2.1. Sampling and preparation

Samples was collected at Dapeng in a biofouling community on a fish farm (GPS coordinate: 22°34’04.2”N 114°31’52.0”E). At the sampling site, samples were obtained by removing a rope covered with intact biofouling community. The rope was sampled as a whole and bought back to the lab. The samples were stored in a opaque box with sea water, and was aerated 3 hours later. The sample was then left overnight. The samples on the rope were then picked up, and individuals of sample was collected, each was taken a photo with a scale bar and ID for recognition.

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After the photo was taken, sample was preserved with 100% ethanol in a 50 mL centrifuge tube.

2.2. DNA extraction

Samples that were previously preserved was dissected. Samples that are larger than 4 cm was dissected and musscle tissues was grinded with porcelain grinder. Samples smaller than 4 cm skipped the step of dissection and was directly grinded and along with shell and exoskeleton for sample that includes them. The grinded individual sample was then transferred to a 2 mL centrifuge tube and samples were added with the lysis solution of the BioFlux's Biospin Marine Animal Genomic DNA Extraction Kit. Samples were then kept in a stirring environment with constant temperature of 50 °C for 3 hours.
After the process of lysis, samples were centrifuged with 12000 g for 3 minutes. 500 μL of supernatant from each sample was then transferred to the complex spin centrifuge tube within the kit. The samples were then added with 700 μL of binding buffer and 300 μL of Anhydrous ethanol. Tubes was then centrifuged again with 10000 g for 1 minutes. The setup of adding buffer and anhydrous and centrifuging was repeated. The spin tube was then centrifuged again without adding the buffers and ethanol, and the columns were transferred to new 2 mL tube. 50 mL of distilled water was then added to the tubes and centrifuged again.
After the centrifution, concentrated and extracted DNA sample was obtained, the exact concentration was measured using Nanodrop. The concentration of solution with the extracted DNA will deterimne the volume of Phanta Master Mix (P525-AA) and primer added. For 500 ng of DNA template, 1μL of LCO1490
(5'- GGTCAACAAATCATAAAGATATTGG-3')
primer and HCO2198
(5'- TAAACTTCAGGGTGACCAAAAAATCA-3')
primer were added and 10 μL of Phanta Master Mix were added. In total the solution is added up to 20 μL.

2.3. Polymerase Chain reaction

The process of PCR amplification includes 4 minutes at 94 °C and then 30 cycles of 30 s at 94 °C, 30 s at 45 °C, 30 s at 72 °C, and finally 7 minutes at 72 °C. Gel electrophoresis was then performed with a marker as compared to the sample and the target DNA sequence was identified by reference to marker. Impure samples have its matching part of DNA sequence cut out from the gel to undergo a second loop of purification. After the process samples were sent for Sanger sequencing.

2.4. Data analysis

The obtained sequence of HCO and LCO amplified COI fragments were trimmed based on sequencing quality, and invalid sequence were removed. The trimmed sequences were then processed by nucleotide BLAST (Blastn) of NCBI. Species identity was determined based on the highest % identification to the matched sequence and its correlation with specimen.

3. Results

3.1. Bacteria isolated from biofilm samples

In total, 20 samples were collected and conducted the PCR test. These samples include a rough classification of 2 barnacles, 5 tube worms, 2 ascidians, 3 polychate, 2 sponge, 2 bivalves and 3 anemones. This classification is roughly based on the sample image takened and will act as part of factor when it comes to comparison between the image and PCR result (Figure 1). From the total 20 samples collected, 14 received a PCR result. The result, along with the status of all 20 samples, is presented in the table below (Figure 1). Each sample received an ID for identification, rough classificaiton and detailed information such as sample location and date (Table 1).
Relationship between foulers and foulers were also recorded. In total, 9 types of fouling pairs were recorded, and in the 20 samples collected, 13 were recorded to be "foulers of fouler". Based on observation, the 9 pairs inculdes:

- Barnacle (DP-1, DP-2) on bivalve
- Tube worm (DP-4, DP-11, DP-20) on encrusting bryozoan
- Ascidian (DP-5, DP-6) on encrusting bryozoan
- Tube worm (DP-10) on sponge
- Sponge (DP-12) on encrusting bryozoan
- Anemone (DP-14) on bivalve
- Bivalve (DP-15) on bivalve
- Anemone (DP-16) on encrusting bryozoan
- Anemone (DP-19) on encrusting bryozoan

Using these samples, DNA extraction and PCR amplification of COI gene fragment were performed, and then DNA sequencing, in order to identify the species identity of each specimen. Out of the 20 samples, the COI fragment of 14 specimens was successfully sequenced and used for species

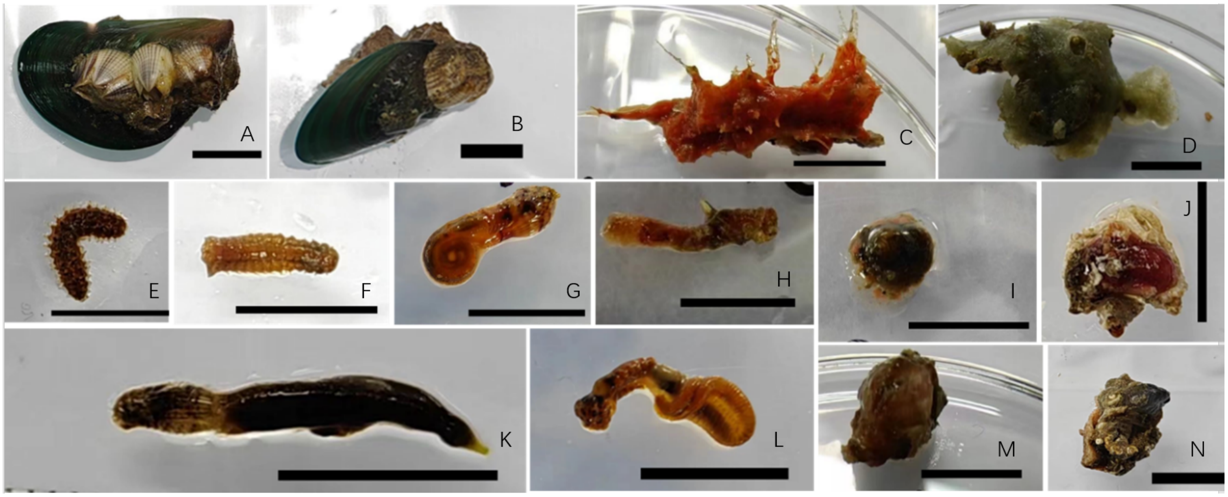

Figure 1. Samples that received a PCR result. Sample A-N represents sample DP-1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, and 14. Black bar represents a length of 10 mm.

Table 1. Information of samples collected from Dapeng.

sample ID	classification (rough)	settle on	COI sequence	Sequence ID	ID %
DP-1	Barnacle	Bivalve	*Amphibalnus reticulatus*	KU204370.1	89.02
DP-2	Barnacle	Bivalve	*Amphibalnus reticulatus*	KU204256.1	89
DP-4	Tube worm	Bryozoan	*Leptodius affinis*	MZ900933.1	99.64
DP-5	Ascidian	Bryozoan	*Lepidonotus squamatus*	AWHO1944.1	89.02
DP-6	Ascidian	Bryozoan	*Leptodius affinis*	MZ900933.1	94.19
DP-7	Polychate	Not fouler	*Lepidonotus carinulatus*	MG193920.1	98.46
DP-8	Polychate	Not fouler	*Rhynchelmis yakimorum*	GU592325.1	76.63
DP-9	Sponge	Artificial surface	*Lepidonotus carnulatus*	MG193920.1	91.46
DP-10	Tube worm	Sponge	*Conchoderma hunteri*	NC_071897.1	98.48
DP-11	Tube worm	Bryozoan	*Leptodius affinis*	MZ900933.1	97.64
DP-12	Sponge	Bryozoan	Unknown		
DP-13	Bivalve	Artificial surface	*Palola* sp.	ABC62016.1	85.54
DP-14	Anemone	Bivalve			
DP-15	Bivalve	Bivalve			
DP-16	Anemone	Bryozoan			
DP-17	Crab	Not fouler			
DP-18	Tube worm	Unknown	*Mancinella echinata*	YP010936444.1	95.65
DP-19	Anemone	Barnacle			
DP-20	Tube worm	Bryozoan			
DP-21	Polychate	Unknown			

identification.
In the 12 samples, 4 samples have its sequence matched to the rough classification. DP-1 and DP-2 was indicated by the COI sequence matching *Amphibalnus reticulatus*, a type of fouling barnacle, which also match the rough classification of barnacle. DP-7 have its COI sequence matching *Lepidonotus squamatus*, which matched the rough classification of polychaete; Same went with DP-8, having its COI sequence matching the polychaete *Rhynchelmis yakimorum*. Both species were considered non-sessile poly-

chaete and is related to biofouling communities. The rest of the sample demonstrated different species. These species were highly different from the rough classification up to the category of classes.

4. Discussion

The result of the 20 samples provided 9 types of fouling relationship base on observation. From this result, several conclusions on the trend of fouling relationship of certain species can be withdrawn.

Among all the samples, sea anemones were observed to foul on other organisms the most often. The sea anemone samples DP-14, 16, and 19 were recorded to be fouling on bivalves (rough classification: mussel), bryozoan, and barnacles. Some species of bivalves are known for their anti-fouling surface of shells [1]. Yet anemone is still able to foul on such a surface. This suggests that anemones have a wide range of selection of fouling surfaces. This can be a consequence of the lifecycle of cnidarians beginning with planktonic larvae and then fouling polyps. This is beneficial to the maintenance of artificial facilities, since as a fouling organism, sea anemone does not form calcified skeletons. This means that sea anemones can compete with foulers that form a calcified skeleton on the ecological niche. Surfaces with reduced calcified skeletons and more sea anemones are easier to remove compared to those with highly calcified skeletons. Such mechanisms, namely, to encourage the settlement of soft or light-calcifiers, have already been applied in the field, often as a method to suppress the growth of the biofouling community [2]. However, this fouling trend may also lead to potential issues when the fouling surface is mobile. Under such conditions, the anemone will have a higher possibility of being an invasive species carried to different places by its fouling surface.

On the other hand, encrusting bryozoans are recorded to have the most samples fouling on (7 samples) Due to its ability to rapidly grow and create calcified skeleton, encrusting bryozoans are known for its aggressive biofouling [3]. While encrusting bryozoans have strong biofouling capability, the factor that they also provide a wide range of other bio-surface to foul on further strengthens its contribution to the growth of biofouling communities. This aspect increases the damage that encrusting bryozoan deal to artificial facilities.

There was also a case where a biofouler was fouling on another biofouler of the same species. Sample DP-15 was a bivalve fouling on another bivalve. This act of fouling on or near the same species may increase the likelyhood for individuals to reproduce sexually, commonly seen in many biofouling communities.

4.1. PCR result

As for the 8 samples that have a difference in rough classification and COI sequence, several theories were proposed to be the partial factor of this phenomenon.

Some species of COI sequence resemble distribution that includes the east Shenzhen coast despite being different from the rough classification based on observation. As an example, DP-5 and DP-11 have a COI sequence matching *Lepidonotus squamatus*, a type of polychaete commonly seen in Asia; This suggests that the result might not be an error, but a misidentification of another species within the biofouling community.

Some of the samples such as sponge and ascidian are well known to be the habitat of many other biofouling and benthic organisms. In samples DP-6 and DP-9, the COI sequence matched a species of crab and a species of polychaete. Both might be dwelling in the sample and as a result the COI fragment of these dwellers were amplified instead of the sponge/ascidian specimen, leading to a mismatch between the photo and the molecular ID of the specimens.

During sample processing, improper dissection and tissue extraction might also lead to the contamination of the PCR result. When the tissue was dissected, other fouling organisms might be contaminated with the tissue. This is especially an issue when it comes to the extraction of small biofouling organisms due to their small size and impurities on their body. In the future, the step of tissue extract might be conducted with a prior washing that can clean a proportion of the impurities.

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